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Heritability and selection of strategic traits in the NFL coaching tree

Anonymised for review

Affiliation removed for anonymous review

Abstract

Human expertise is transmitted socially, and where learners have identifiable teachers the transmission leaves a genealogy. We treat the National Football League coaching tree as a cultural pedigree and, for a preregistered set of ten standardized play-calling traits, estimate each trait's transmissibility (h^2 , mentor-to-protege resemblance from a Bayesian parent-offspring model with a measurement-error correction) and its selection (S , association with a coach's wins above replacement). Every trait is a genuine, repeatable individual characteristic (repeatabilities 0.34 to 0.58), yet transmissibility and selection come apart. Schematic-identity traits transmit but do not win: shotgun $h^2 = 0.45$ (95% HDI [0.27, 0.65]) with $S = 0.08$, and tempo similarly. Offensive aggression is the mirror image, repeatable (0.40) and selected ($S = 0.18$, [0.08, 0.27]) yet not transmitted ($h^2 = -0.15$); a preregistered contrast confirms it is repeatable but not heritable (0.55, [0.24, 0.87]). Defensive aggression alone both transmits ($h^2 = 0.49$) and wins ($S = 0.15$). The predicted negative heritability-fitness relationship does not appear (Spearman +0.32, [-0.54, 0.69]): rather than a trade-off, transmission and selection are largely decoupled. Coaching lineages propagate systems, not success, and the fitness-linked behavior spreads through a channel a purely vertical lens would miss.

Keywords: cultural evolution, social learning, vertical transmission, heritability, cultural fitness, sport

1. Introduction

Human skilled practice is transmitted socially, and where practitioners learn from identifiable teachers the transmission leaves a genealogy. Coaching in the National Football League (NFL) is an unusually legible case. Head coaches hire assistants, assistants become coordinators, coordinators are promoted to head coach, and the record of who worked under whom is documented season by season. The result is a “coaching tree”: a directed network of mentor–protege ties along which styles of play are widely believed to descend. Popular accounts speak of the Walsh tree or the Belichick tree much as a breeder speaks of a bloodline, and NFL hiring leans heavily on that pedigree logic, on the assumption that working under a successful coach transfers both winning approaches and strategic expertise.

Cultural evolution treats socially learned behavior as a second inheritance system, one that, like the genetic system, is transmitted, varies, and is subject to selection (Boyd & Richerson, 1985; Cavalli-Sforza & Feldman, 1981; Henrich, 2016). Much of the human capital that distinguishes experts, from surgeons to craftspeople to coaches, is acquired by working under a more accomplished practitioner rather than from books, so expertise descends along chains of apprenticeship. Measuring vertical transmission in such chains is nonetheless hard: it requires observing the same behavior in both teacher and learner, on a comparable scale, with the genealogy known, and human pedigrees of behavior are seldom that legible. Professional sport is an exception, because the record of who coached under whom is public and the behavior itself, every play call, is logged.

Recent work has begun to treat sport tactics as a cultural-evolutionary system. Mesoudi (2020), in this journal, analyzed the cultural evolution of association-football formations and showed that a

managerial innovation spread through strategic social learning among contemporaries, a *horizontal* diffusion that accelerated as the tactic proved successful. That study measured transmission among peers and explicitly named transmission along mentor–protege lineages, the *vertical* channel, as open work. The present paper takes up exactly that question, for a different sport. Cultural transmission theory distinguishes vertical transmission, from a parent-like model to a learner, from horizontal transmission among contemporaries, and the two need not carry the same traits (Boyd & Richerson, 1985; Cavalli-Sforza & Feldman, 1981). A coaching tree is an unusually direct substrate for the vertical channel, because both the mentor’s and the protege’s on-field behavior are observed and dated. As we will show, the vertical and horizontal channels in coaching turn out to move *different* strategic traits, so the two studies are complementary halves of one picture.

To ask how faithfully behavior descends such a pedigree, we borrow the toolkit quantitative genetics built for that decomposition. Two quantities anchor it. Repeatability is the share of a trait’s variance that is a stable property of the individual rather than occasion-to-occasion noise, and it sets an upper bound on how much of the trait can be transmitted at all. Heritability is the share actually passed to descendants; in a parent–offspring design it is read directly from the resemblance between parent and offspring phenotypes, with no need for an assumed relatedness matrix (Falconer & Mackay, 1996). The same variance-partitioning logic extends from classical pedigrees to repeated measures through the mixed-model animal model (Nakagawa & Schielzeth, 2010; Wilson et al., 2010), and dual-inheritance theory has long held that it applies to culturally transmitted traits as well as genetic ones (Cavalli-Sforza & Feldman, 1981). What has been missing for culture is a system in which both the pedigree and the phenotype are observed directly rather than inferred; the coaching tree supplies one.

We use these tools to place each trait on two axes. The first is transmissibility, the cultural analog of heritability h^2 : how strongly a protege’s own head-coaching style resembles the style of the coach they apprenticed under. The second is selection S : how strongly a trait covaries with success, measured here as a coach’s context-adjusted wins above replacement (WAR). In biological populations the two are famously in tension. Fisher’s fundamental theorem implies that selection consumes the additive genetic variance of the traits most closely tied to fitness, so those traits should be the *least* heritable (Fisher, 1930). The prediction holds across taxa: Mousseau and Roff (1987) found life-history traits, those nearest fitness, markedly less heritable than morphological ones, and Kruuk et al. (2000) report a rank correlation near -0.8 between the heritability of a suite of red-deer traits and their association with fitness. Whether an analogous tension arises in a cultural system, where inheritance is social learning rather than meiosis and the variance-depleting logic of Fisher’s theorem need not apply, is unknown.

The coaching tree is a favorable model system for the question. Every offensive and defensive decision is recorded play by play, so a coach’s tendencies can be measured as deviations from what the game situation demands; the mentor–protege genealogy is documented, so vertical transmission can be traced from a coach’s coordinator apprenticeship to their own head-coaching tenure; and coaching success has a context-adjusted measure, wins above replacement, that separates a coach’s contribution from roster talent. Because a coordinator’s later head-coaching behavior is observed after and apart from the mentor’s, the design is a genuine between-person parent–offspring comparison rather than a coach correlated with himself.

We ask a single organizing question: in the NFL coaching tree, are the strategic traits that transmit the same traits that win? We separate strategy into two families. *Approach-to-scenarios* traits describe how aggressively a coach attacks game situations (going for it on fourth down, passing in neutral situations, throwing beyond the first-down marker, attempting two-point conversions, and on defense loading the box, sending extra rushers, and playing man coverage). *Schematic-identity* traits describe the formational signature of a team (how much it operates from shotgun, and how fast it plays). The split is cleanest on offense; as we find, defensive “aggression” mixes an attitudinal and a

schematic component (Discussion). Our preregistered hypotheses were:

- H1. Approach traits are more strongly *selected* (predict WAR), while identity traits are more strongly *inherited* (transmit down the tree).
- H2. Offensive aggression is a repeatable individual trait that is nonetheless not heritable: its within-coach repeatability clearly exceeds its transmissibility.
- H3. Across the ten sub-traits, heritability and selection are negatively related: the traits that transmit most strongly are not the ones that predict winning.

These predictions were registered before any confirmatory quantity was computed on the final measurement pipeline (Section 2). As reported below, the data support a decoupling of transmission and selection but not the specific negative trade-off of H3, and the picture is enriched by one trait, defensive aggression, that both transmits and wins.

2. Materials and methods

2.1 Coaching phenotypes (“genes”)

We construct standardized coach-season phenotypes from play-by-play data for the 2006–2024 NFL seasons (obtained through `nfl_data_py`; more than 600,000 plays). For each strategic decision we train a play-level expectation model (gradient-boosted trees) on pre-play game context only, deliberately excluding season, week, and coach identity, so the baseline reflects what an average coach would do in identical circumstances and is time-invariant. A coach-season phenotype is the average deviation of observed behavior from this expectation. Writing a_{cj} for coach c ’s action on play j (an indicator for the discrete decisions, a continuous value for pace and the box and rusher counts) and $\hat{a}_{cj}$ for the model’s expectation given that play’s pre-snap context, the raw gene is

$$g_c = \frac{1}{n_c} \sum_{j=1}^{n_c} a_{cj} - \frac{1}{n_c} \sum_{j=1}^{n_c} \hat{a}_{cj}, \quad (1)$$

and the phenotype is its standardization across coach-seasons, $\tilde{g}_c = (g_c - \bar{g})/\text{sd}(g)$; positive values denote more aggressive, faster, or more shotgun-oriented play than context predicts. This residual construction isolates a strategic tendency from forced circumstance: a coach who goes for it often only because he is always trailing is not thereby aggressive.

The ten sub-traits are the residuals of ten play-level models (Table 1): four offensive-aggression decisions (going for it on fourth down, passing versus running in neutral situations, targeting beyond versus short of the first-down marker, and attempting a two-point conversion), two tempo behaviors (no-huddle use and seconds between snaps), three defensive behaviors (defenders in the box, number of pass rushers, and man versus zone coverage), and shotgun formation use. Each is a gradient-boosted model on pre-play context (down, distance, field position, score, clock, and environment), a classifier for the discrete decisions and a regressor for the continuous ones (pace, box count, rushers). Situational context explains most of the variance in each decision, leaving the smaller, coach-specific residual we read as tendency; each model’s type and out-of-sample performance are given in Table 1. The expectation models, the leakage controls that keep a coach’s own tendencies out of the yardstick that measures them, and the reliability weighting are detailed in the next two subsections.

Every play is attributed to the head coach who led the team for that game, so a within-season head-coaching change splits the season between the two coaches at the game of the change. Coordinators are not assigned their own phenotype; the phenotype is defined only at the head-coach level, so a play is attributed to one coach and never to both a head coach and a coordinator at once. To keep a coach’s own tendencies out of the yardstick that measures them, expectation models are fit with the scoring coach held out and with cross-fitting folds balanced across eras, so each play is scored by a model that never trained on that coach.

Table 1. The ten frozen sub-traits: gene-level composite and family, the play-level expectation model whose residual defines the phenotype and its out-of-sample performance (AUC for classifiers, R^2 for regressors, under grouped resampling that holds out whole team-years), and the data window. All transitions tested are offensive-coordinator-to-head-coach except the three defensive traits, which are defensive-coordinator-to-head-coach.

Sub-trait	Composite (family)	Model	Perf.	Window
Fourth-down	Offensive aggression (approach)	classifier	AUC 0.98	2006–2024
Pass-heavy	Offensive aggression (approach)	classifier	AUC 0.78	2006–2024
Deep-pass	Offensive aggression (approach)	classifier	AUC 0.73	2006–2024
Two-point	Offensive aggression (approach)	classifier	AUC 0.93	2006–2024
No-huddle	Tempo (identity)	classifier	AUC 0.87	2006–2024
Pace	Tempo (identity)	regressor	R^2 0.43	2006–2024
Box-stacking	Defensive aggression (approach)	regressor	R^2 0.43	2016–2024
Pass-rush	Defensive aggression (approach)	regressor	R^2 0.06	2016–2024
Man-coverage	Defensive aggression (approach)	classifier	AUC 0.71	2018–2024
Shotgun	Shotgun (identity)	classifier	AUC 0.82	2006–2024

Ten sub-traits are frozen for the primary analysis (Table 1), grouped into four gene-level composites (offensive aggression, tempo, defensive aggression, and shotgun) that serve as labeled exemplars but are not counted as independent points in the cross-trait test.

2.2 Expectation models, leakage control, and feature selection

Each of the ten expectation models is a gradient-boosted tree ensemble, consistently the strongest model class for tabular, point-in-time prediction of this kind (Chen & Guestrin, 2016); the residual definition of a gene is identical whether the model is a classifier or a regressor. Three safeguards make the residuals trustworthy as coach phenotypes.

First, leakage-free scoring. Because a gene is a residual against a model’s prediction, it is meaningful only if a coach-season is scored in exactly the feature space the model was trained in. We fit every preprocessing step, the categorical encoders, the median imputer, and a truncated-SVD reduction, on the training split alone, persist them, and apply them in transform-only mode when scoring the full play-by-play record; this removes a train/serve mismatch that would otherwise place genes in a different space than the one the models learned. Coach identity, season, and week are excluded as predictors, so no model can recover a coach from a name or the calendar, and every split is made at the team-year group level so that no team or coach appears in both training and evaluation.

Second, coach-held-out cross-fitting. A model trained on a coach’s own plays would partly fit that coach’s tendencies and shrink the very residual we want to measure. The predictions used to compute a gene are therefore out-of-fold, from a group k -fold that leaves out whole coaches (offensive genes) or whole coach-defenses (defensive genes), reusing the tuned hyperparameters, with folds balanced across eras (stratified by career midpoint). Each coach-season is scored by a model that never saw that coach, so the residual reflects deviation from a coach-blind baseline. Model performance (Table 1) is estimated under the same grouped resampling, with Nadeau–Bengio corrected resampled intervals (Nadeau & Bengio, 2003) that account for the overlap among resampled training sets.

Third, feature selection and calibration. Features are not hand-curated. A pre-specified core of fine-grained game-state fields (down, distance, field position, score margin, clock, and time-outs) is always retained; the remaining context variables (Table 2) are chosen by stability selection (Meinshausen & Bühlmann, 2010). Over B subsamples of team-year groups, letting $\hat{S}_K^{(b)}$ be the K

Table 2. Candidate feature pool for the expectation models. The game-state core is always retained; the remaining categories are subject to stability selection. Offensive models draw on up to 26 candidates (21 for two-point, which omits field position); defensive models add the two observed pre-snap indicators.

Category	Fields
Game state (core)	down, distance, yard line, score differential, quarter, seconds remaining, timeouts (both teams)
Field position	goal-to-go, side of field
Environment	home/away, division game, roof, surface, temperature, wind
Drive context	drive play count, drive first downs, net yards
Betting market	spread, total
Pre-snap (defensive models)	observed shotgun, observed no-huddle

highest-importance features in subsample b , the selection frequency of feature k is

$$\hat{\Pi}_k^{(K)} = \frac{1}{B} \sum_{b=1}^B \mathbf{1} \left\{ k \in \hat{S}_K^{(b)} \right\}, \quad (2)$$

and we retain features with $\hat{\Pi}_k^{(K)} \geq \pi$ at threshold $\pi = 0.6$ (taking the union over shortlist lengths K), which prunes collinear inputs without materially changing accuracy. Because the classifier genes are residuals from predicted probabilities, we confirmed the classifiers are well-calibrated out-of-fold (expected calibration error at most 0.03), so the residuals are not artifacts of miscalibration.

2.3 Contemporary-group (era) adjustment

Each phenotype is expressed as a deviation from its own season's league mean (within-season demeaning) before any modeling. Because both a mentor's and a protege's values are re-centered on their own seasons, league-wide drift is removed exactly once and is not counted as transmission or selection. This is the single era control, the cultural analog of the contemporary-group (herd-year-season) adjustment used to estimate breeding values in the animal model; no further season or era term enters the models. Coach WAR is already a within-season, replacement-relative quantity (about 3% between-season variance) and receives no further adjustment.

2.4 Reliability and measurement error

Each coach-season phenotype is an estimate with a known sampling variance. For a probability sub-trait, a coach-season residual θ_i has sampling variance $\nu_i = \sum \hat{p}(1 - \hat{p})/n_i^2$ over its n_i plays (with the analogous residual-variance form for the regression sub-traits); we estimate the between-coach-season variance $\hat{\tau}^2$ with the DerSimonian–Laird random-effects estimator (DerSimonian & Laird, 1986). Each coach-season enters its composite with weight equal to its reliability,

$$\text{rel}_i = \frac{\hat{\tau}^2}{\hat{\tau}^2 + \nu_i}, \quad (3)$$

so a component measured from few plays contributes less; the single-component shotgun gene uses the analogous empirical-Bayes shrinkage toward the league mean. The same per-observation sampling variance, rescaled to the standardized scale, is carried explicitly as a measurement-error level in the heritability (C1) and repeatability (C2) models, so both are estimated for the latent trait rather than its noisy observation, and composites propagate their components' sampling variance. Reliabilities vary across sub-traits (for example the offensive-aggression components run from about 0.65 for fourth-down to 0.91 for pass-heavy), which is why carrying measurement error, rather than treating every phenotype as equally precise, matters for the cross-trait comparison in C3.

2.5 C1: transmissibility (h^2)

A coordinator-to-head-coach transition pairs two different coaches. The outcome is the protege's own phenotype once they became a head coach; the predictor is the phenotype of the head coach they apprenticed under, measured on that mentor's team over the seasons the protege served there as a coordinator. Each coach contributes at most one transition per coordinator role (the most recent coordinator stint preceding a head-coaching stint), which avoids pseudo-replication. Because the phenotype is head-coach-level, this design measures whether the head-coaching style a coach apprenticed under predicts that coach's own later style; it is a between-person parent-offspring comparison, not a coordinator's own trait.

We estimate h^2 with a Bayesian multilevel parent-offspring model fit at the coach-season grain (Nakagawa & Schielzeth, 2010; Wilson et al., 2010). For protege i , a latent mentor level m_i and a latent own level a_i are linked by

$$a_i = \alpha + \beta m_i + \varepsilon_i, \quad \varepsilon_i \sim \text{Normal}(0, \sigma_a), \quad (4)$$

and each individual mentor-overlap season x_{ij} and protege head-coaching season y_{ik} loads on the corresponding latent level with a variance that combines a shared true within-coach season-to-season component τ^2 and the known measurement variance of that observation:

$$x_{ij} \sim \text{Normal}(m_i, \sqrt{\tau^2 + \text{se}_{ij}^2}), \quad y_{ik} \sim \text{Normal}(a_i, \sqrt{\tau^2 + \text{se}_{ik}^2}). \quad (5)$$

Modeling individual seasons rather than pre-averaging lets the mentor predictor enter as a latent variable carrying its measurement variance, an errors-in-variables correction for the attenuation that predictor noise would otherwise cause. The transmissibility is the parent-offspring regression slope at the coach-season grain,

$$h^2 = \frac{\beta \sigma_m^2}{\sigma_m^2 + \tau^2} = \beta_{\text{latent}} \times \text{repeatability}, \quad (6)$$

which is bounded above by the mentor repeatability $\sigma_m^2/(\sigma_m^2 + \tau^2)$: a trait cannot transmit more faithfully than it is a stable trait to begin with. Because a mentor transmits a whole style rather than half a genome, the slope itself is the transmissibility, with no Mendelian doubling. Priors are $\text{Normal}(0, 1)$ on the intercept and slope (phenotypes are z-scored) and half- $\text{Normal}(0, 1)$ on the variance components; sampling uses four chains with 2000 post-warmup draws each, a non-centered parameterization, and target diagnostics $\hat{R} \leq 1.01$ and bulk effective sample size > 400 (met for every fit). As robustness checks the slope is refit with a mentor varying intercept and, in a frequentist version, with mentor-clustered standard errors.

2.6 C2: repeatability

Per-trait repeatability is a Bayesian variance-components intraclass correlation on the full head-coach coach-season panel, with the per-season measurement variance carried as an explicit level so the ratio is the latent-trait repeatability, $\sigma_{\text{between}}^2/(\sigma_{\text{between}}^2 + \sigma_{\text{within}}^2)$. It is the ceiling for h^2 ; because heritability cannot exceed repeatability, we report $P(h^2 > \text{repeatability})$ per trait, a value near one flagging a specification problem rather than a substantive result.

2.7 Selection (S)

Coach WAR is our fitness proxy, and it is chosen deliberately. It estimates the wins a head coach adds relative to a replacement-level alternative, adjusting for roster talent, schedule, and game context, so it credits the coaching contribution rather than a strong roster; it is validated out of sample on

held-out coaches in companion work (Anonymous, 2026). Being replacement-relative and computed within each season, it is a relative-fitness analog centered near zero (median approximately zero) and nearly era-flat (about 3% between-season variance), so, unlike the genes, it needs no separate era adjustment. WAR is taken from a version-pinned companion estimator; if that estimator is revised, S is recomputed on the revised version as a pre-specified robustness check.

Selection S is then the era-adjusted, inverse-variance-weighted correlation of a trait with Coach WAR: within-season demean the standardized gene and WAR, weight each coach-season by its games-based WAR precision, and take the coach-clustered weighted correlation. Single-season WAR is noisy, and because mean-zero noise in the outcome attenuates a correlation rather than inflating it, this estimate is a conservative floor.

2.8 C3 and decision rules

For the ten sub-traits we assemble the pair (h^2, S) and compute their Spearman rank correlation with a 95% bootstrap interval that resamples the traits and draws h^2 from its posterior and S from its bootstrap. The rank form is the established test of the heritability-fitness relationship in quantitative genetics (Kruuk et al., 2000; Mousseau & Roff, 1987). Each hypothesis is judged by one pre-specified interval rule. H1 is supported if the approach family's mean S exceeds the identity family's and the identity family's mean h^2 exceeds the approach family's, each difference with a 95% interval excluding zero. H2 is supported if (aggression repeatability minus aggression h^2) has a 95% interval excluding zero. H3 is supported if the Spearman correlation is negative with a 95% bootstrap interval excluding zero.

2.9 Exploratory analyses

Two analyses are declared exploratory and corrected among themselves under a Benjamini-Hochberg false-discovery rate (Benjamini & Hochberg, 1995), separately from the confirmatory tests. The first is a within-era Moran's I on the mentor network (an undirected, row-normalized adjacency of mentor-protégé ties, with a permutation null that shuffles trait labels only within career-midpoint era blocks) (Moran, 1950). The second is reproductive fitness as an exposure-normalized rate: head-coaching offspring (protégés who themselves became head coaches) per head-coaching season, with a minimum-exposure floor, together with the test of whether coach quality predicts that rate net of exposure.

2.10 Software

Analyses are in Python; the Bayesian models use PyMC. The pipeline is seed-fixed and runs from a single command.

3. Results

3.1 The phenotypes are repeatable coach traits

Before asking how a trait transmits or whether it wins, we confirm it is a stable individual characteristic rather than a team artifact. Every trait is repeatable: the latent-trait repeatabilities (Table 4) range from 0.34 (pass-heavy) to 0.58 (fourth-down), all clearly above zero, so each phenotype carries substantial stable between-coach signal net of measurement noise. This is the quantitative-genetics prerequisite for a meaningful heritability: a trait must be repeatable before it can be transmissible.

Three companion analyses corroborate that the genes belong to the coach rather than the team or its roster (Table 3). First, a one-way variance decomposition attributes far more of each gene's variance to coach identity than to team identity (for composite offensive aggression, 0.50 versus 0.14). Because η^2 inflates mechanically with the number of groups, and a coach maps to far more levels than a team, we benchmark each against a cardinality-matched label-permutation null: the coach excess

Table 3. The offensive genes are coach traits. One-way variance shares (coach, team, season; marginal η^2 , which need not sum to one because coach and team are confounded), cross-team travel (era-adjusted across-team correlation among the 32 movers, 95% CI), and year-to-year persistence (lag-1 correlation between consecutive coach-seasons; computed for the aggression genes).

Gene	coach η^2	team η^2	season η^2	travel r [95% CI]	persist. r
Composite offensive aggression	0.50	0.14	0.18	0.55 [0.29, 0.73]	0.50
fourth-down	0.51	0.07	0.26	0.09 [−0.24, 0.42]	0.53
pass-heavy	0.49	0.18	0.08	0.32 [−0.03, 0.60]	0.55
deep-pass	0.44	0.12	0.05	0.17 [−0.20, 0.51]	0.46
two-point	0.44	0.09	0.17	0.30 [0.04, 0.60]	0.43
Shotgun	0.57	0.12	0.48	0.28 [−0.18, 0.68]	–
Tempo	0.52	0.11	0.07	0.48 [−0.11, 0.82]	–

over its null is about 0.29 against 0.08 for team, so the coach signal is roughly three times larger, and the same ordering holds for every gene. Second, when coaches change teams their tendencies travel with them: across the 32 coaches who moved, composite offensive aggression at the new team tracks the gene they showed at the old one ($r = 0.55$, 95% CI [0.29, 0.73], on the era-adjusted gene), and in a mover regression the coach's prior-team gene carries $\beta = 0.54$ ([0.26, 0.88]) net of the destination's pre-arrival identity. Third, the genes persist from season to season (composite offensive aggression lag-1 $r = 0.50$, with its four components between 0.43 and 0.55). At only 32 movers the travel test is imprecise for any single gene, which is why the composite, which averages out component sampling noise, travels most cleanly.

Together with the repeatabilities, these lines establish that the phenotypes are stable properties of the coach rather than artifacts of a roster, the prerequisite for treating them as heritable-eligible traits.

3.2 Transmissibility and selection come apart

Table 4 reports, for each trait, the preregistered transmissibility h^2 (C1), repeatability (C2), and selection S . Two facts stand out. First, transmissibility varies sharply across traits that are all similarly repeatable: the schematic-identity genes transmit strongly (shotgun $h^2 = 0.45$, 95% HDI [0.27, 0.65]; no-huddle 0.30, [0.09, 0.51]) as does defensive aggression (0.49, [0.13, 0.85]), while offensive aggression does not transmit at all (−0.15, [−0.46, 0.12]). Second, selection follows a nearly opposite ordering: the approach components predict WAR (offensive aggression $S = 0.18$, [0.08, 0.27]; defensive aggression 0.15, [0.03, 0.27]; pass-heavy 0.16; pass-rush 0.17) while the identity genes do not (shotgun 0.08, [−0.03, 0.17]; tempo 0.03). The frequentist mentor-clustered slopes and the mentor varying-intercept refits agree with the Bayesian h^2 (for example shotgun clustered slope 0.47, $p < 0.001$; defensive 0.61, $p = 0.03$; offensive aggression −0.03, $p = 0.85$). Figure 1 plots each trait's h^2 against its repeatability ceiling and makes the pattern legible at a glance.

3.3 The heritability by selection map and the preregistered verdicts

Placing the four composites in $h^2 \times S$ space (Figure 2) makes the structure visible. Offensive aggression sits in the adaptive-but-personal corner (selected, not heritable); shotgun and tempo sit in the heritable-but-neutral corner (transmitted, not selected); defensive aggression sits alone in the adaptive-and-heritable corner. The two axes are close to orthogonal.

The three preregistered decision rules resolve as follows.

H2 is supported. Offensive aggression is repeatable (0.40) but not heritable ($h^2 = -0.15$); the preregistered contrast, repeatability minus h^2 , is 0.55 with a 95% HDI of [0.24, 0.87], excluding zero, and $P(h^2 > \text{repeatability}) = 0.00$. Offensive aggression is a real, stable, personal trait that

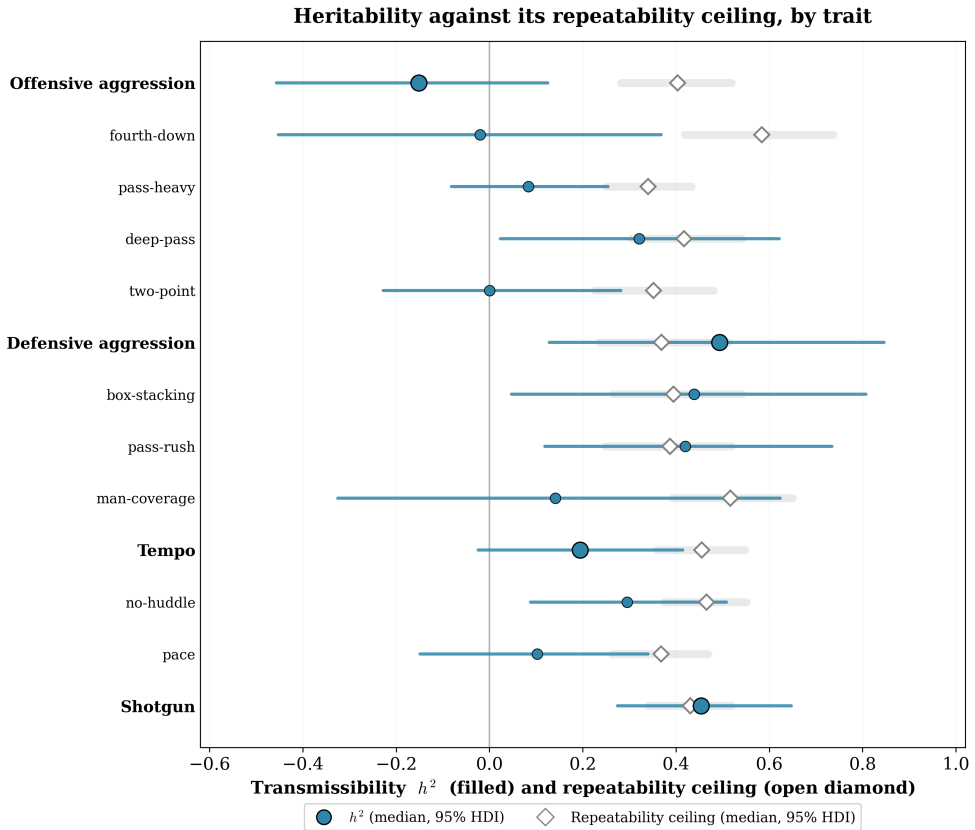


Figure 1. Transmissibility (h^2 , filled points with 95% HDIs) against its repeatability ceiling (open diamond, median; gray band, repeatability 95% HDI) for every trait, grouped by gene and colored by family. A heritability cannot exceed its repeatability in the population; because h^2 and repeatability are estimated from separate models with wide posteriors, their medians can cross for traits that transmit near the ceiling, but every h^2 point falls within the repeatability's credible interval (the posterior contrast $P(h^2 > \text{repeatability})$ is 0.5 to 0.75 for these, that is $h^2 \approx \text{repeatability}$). Offensive aggression and its components are repeatable but have h^2 indistinguishable from zero, a clean gap ($P = 0.00$), the preregistered repeatable-but-not-heritable pattern (H2); schematic identity and the defensive front transmit near their ceiling.

does not pass down the tree. The contrast behaves as the prereg intended elsewhere too: for the traits that transmit near their ceiling (shotgun, defensive aggression, box-stacking, pass-rush) h^2 and repeatability are statistically indistinguishable ($P(h^2 > \text{repeatability})$ between 0.5 and 0.75, each h^2 inside the repeatability's credible interval), so the apparent crossings in Figure 1 are estimation noise near the ceiling, not violations of the bound.

H1 is not supported as a conjunction, though its selection half holds. The approach family is more selected than the identity family (mean- S difference +0.11, 95% interval [0.01, 0.22], excluding zero), but the identity family is not reliably more heritable than the approach family (mean- h^2 difference +0.16, [-0.12, 0.43], including zero). The heritability half fails for a substantive reason: defensive aggression is an approach trait that is nonetheless highly heritable ($h^2 = 0.49$), so the approach family does not have uniformly low transmissibility.

H3 is not supported. Across the ten sub-traits the Spearman correlation between h^2 and S is +0.32 with a 95% bootstrap interval of [-0.54, 0.69] (Figure 3): the point estimate is positive, the opposite of the predicted negative trade-off, and the interval spans zero. The biological heritability-fitness trade-off does not replicate here.

Table 4. Transmissibility (h^2 , C1), repeatability (C2), and selection (S) for the four composites and the ten sub-traits. h^2 and repeatability are posterior medians with 95% HDIs; S is the era-adjusted inverse-variance-weighted correlation with WAR with a 95% coach bootstrap CI. All C1/C2 fits met $\hat{R} \leq 1.01$ and bulk ESS > 400.

Trait	h^2	repeatability	S
<i>Offensive aggression</i>	−0.15 [−0.46, 0.12]	0.40 [0.28, 0.52]	+0.18 [0.08, 0.27]
fourth-down	−0.02 [−0.45, 0.37]	0.58 [0.42, 0.74]	+0.04 [−0.05, 0.13]
pass-heavy	+0.08 [−0.08, 0.25]	0.34 [0.25, 0.43]	+0.16 [0.04, 0.27]
deep-pass	+0.32 [0.02, 0.62]	0.42 [0.30, 0.54]	+0.09 [0.00, 0.17]
two-point	+0.00 [−0.23, 0.28]	0.35 [0.23, 0.48]	−0.02 [−0.11, 0.07]
<i>Tempo</i>	+0.19 [−0.02, 0.41]	0.45 [0.36, 0.55]	+0.03 [−0.07, 0.14]
no-huddle	+0.30 [0.09, 0.51]	0.47 [0.37, 0.55]	−0.02 [−0.11, 0.08]
pace	+0.10 [−0.15, 0.34]	0.37 [0.26, 0.47]	+0.07 [−0.04, 0.19]
<i>Defensive aggression</i>	+0.49 [0.13, 0.85]	0.37 [0.24, 0.51]	+0.15 [0.03, 0.27]
box-stacking	+0.44 [0.05, 0.81]	0.39 [0.26, 0.54]	+0.06 [−0.04, 0.16]
pass-rush	+0.42 [0.12, 0.73]	0.39 [0.25, 0.52]	+0.17 [0.04, 0.29]
man-coverage	+0.14 [−0.33, 0.62]	0.52 [0.39, 0.65]	+0.13 [−0.01, 0.28]
<i>Shotgun</i>	+0.45 [0.27, 0.65]	0.43 [0.34, 0.52]	+0.08 [−0.03, 0.17]

Taken together, the verdicts describe a *decoupling* rather than a trade-off. Transmission and selection are close to independent across coaching traits, with two clean archetypes, inherited-but-neutral identity (shotgun, tempo) and selected-but-personal offensive aggression, and one trait, defensive aggression, that occupies the both corner and is chiefly responsible for the absence of a clean negative relationship.

3.4 Exploratory: network signal and reproductive success

Two exploratory analyses, corrected among themselves under a Benjamini–Hochberg false-discovery rate, round out the picture; neither is confirmatory.

First, phylogenetic signal on the mentor network (Table 5, Supplementary material). Moran’s I , the network analog of the tendency of related individuals to resemble one another, is positive for the schematic and defensive genes and essentially zero for offensive aggression (defensive aggression $I = 0.21$, within-era permutation $p = 0.002$; tempo 0.12 , $p = 0.008$; shotgun 0.08 , $p = 0.04$; offensive aggression -0.01 , $p = 0.52$). Under the exploratory correction the defensive and tempo signals survive. The pattern mirrors the transmission results exactly: network-adjacent coaches resemble one another in the traits that transmit and not in the trait that does not, a second and independent line of evidence for the same decoupling.

Second, reproductive success, the other face of fitness (Figure 4). Counting a mentor’s distinct proteges who themselves became head coaches, the distribution is heavy-tailed: across 463 head coaches the Gini coefficient is 0.66 and the most prolific decile produced 43% of all head-coaching descendants, with Marty Schottenheimer and Bill Parcells (13 each), Bill Belichick (12), and Andy Reid (11) at the top. Reproductive success tracks career WAR ($r = 0.36$), but the link runs through longevity rather than per-season quality: offspring count correlates 0.71 with seasons coached, and net of tenure the exposure-normalized rate (offspring per head-coaching season) is essentially unrelated to quality (partial $r = 0.09$). A coach’s ecological fitness (winning) and reproductive fitness (descendants) are thus only weakly linked, and through survival: coaches shape the league less by winning in any given season than by lasting long enough to send many proteges into head-coaching chairs, carrying their schematic identity, though not their competitive edge, with them.

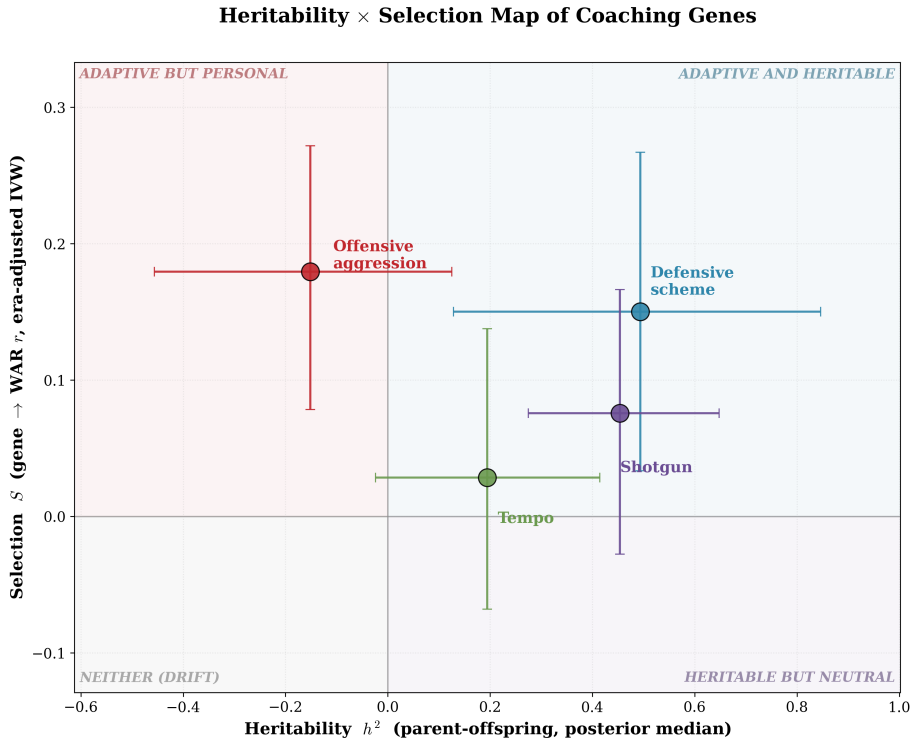


Figure 2. Heritability (h^2 , C1 posterior median) versus selection (S , era-adjusted inverse-variance-weighted gene-WAR correlation) for the four gene-level composites, with 95% intervals on both axes. Offensive aggression is adaptive but personal; shotgun and tempo are heritable but neutral; defensive aggression is both. The axes are nearly independent.

4. Discussion

4.1 Decoupling, not a trade-off

We set out to test whether the traits that transmit down a coaching lineage are the traits that win, expecting the negative heritability–fitness relationship familiar from biological populations. That specific trade-off did not appear. Instead, transmission and selection are largely *decoupled*: across ten strategic traits the two axes are close to orthogonal (cross-trait Spearman +0.32, interval spanning zero), not negatively related. The result is not the absence of structure but a different structure. Two archetypes are clean. Schematic-identity traits, shotgun and tempo, are transmitted faithfully (h^2 up to 0.45) yet carry no detectable fitness consequence ($S \approx 0$); in evolutionary terms they behave like heritable but selectively neutral markers (Kimura, 1968), tags of lineage that ride along with apprenticeship rather than with winning. Offensive aggression is the mirror image, a repeatable and selected but non-transmitted trait (h^2 indistinguishable from zero, $S = 0.18$): what wins on offense is a personal characteristic that does not pass down. This last pattern is the preregistered H2, and it is the sharpest single result: a trait can be a stable individual property (repeatability 0.40) and predict success and still not be inherited.

We preregistered a negative relationship and did not find one, and we report that plainly; the value of registering H1–H3 in advance is that the null on the trade-off is a result rather than a quiet omission. It also invites a sharper reading through the breeder’s equation, $R = h^2 S$, which gives a trait’s per-generation response to selection along a lineage. For offensive aggression the lineage prediction is essentially nil ($R \approx h^2 S \approx 0 \times 0.18 = 0$), because the trait is not transmitted, and yet its league-wide mean rose steadily over our window. A quantity that the breeder’s equation says should

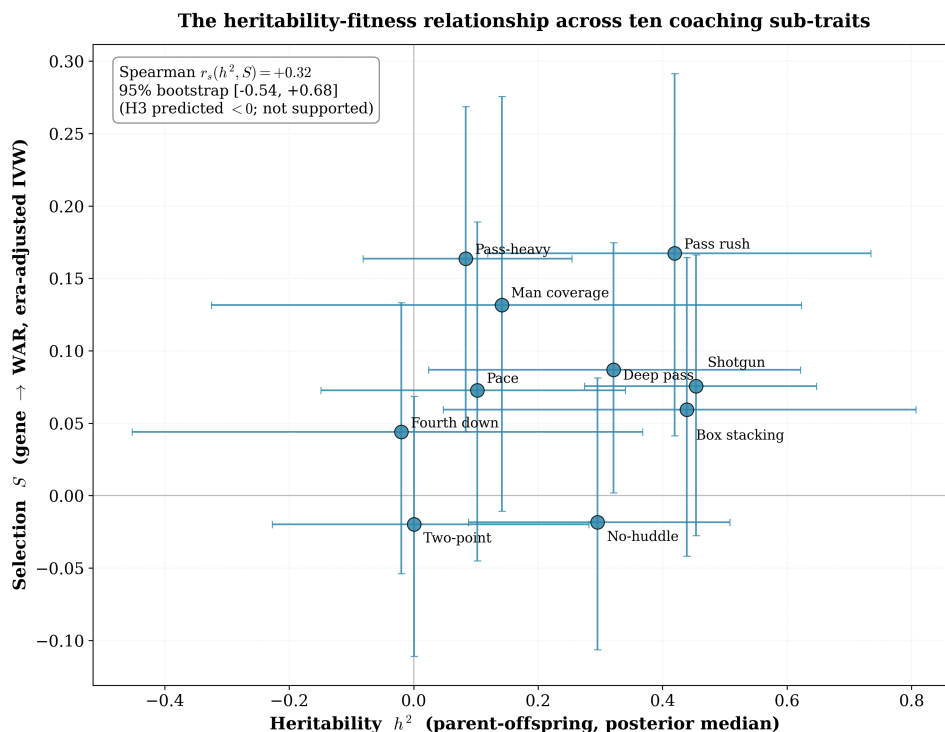


Figure 3. The cross-trait heritability-fitness relationship, the cultural analog of the wild-population test of Mousseau and Roff (1987) and Kruuk et al. (2000): each of the ten sub-traits placed by transmissibility h^2 and selection S , with 95% intervals (a single colour throughout; the approach-versus-identity contrast is drawn in the text, not by segmenting the figure). The Spearman rank correlation (H3) is +0.32 ([−0.54, 0.69]), not the predicted negative, so the two axes are decoupled rather than traded off.

barely change down the tree changed markedly across the population, the empirical signature that a second, non-vertical channel is at work.

4.2 Two channels: vertical down the tree, horizontal across the league

The classical trade-off assumes a single inheritance channel along which selection erodes heritable variance. Coaching has two (Boyd & Richerson, 1985; Cavalli-Sforza & Feldman, 1981). Schematic identity spreads *vertically*, down the offensive-coordinator apprenticeship: it carries the high h^2 and the positive phylogenetic signal (Table 5, Supplementary material) that vertical transmission produces. Offensive aggression, lacking a vertical channel ($h^2 \approx 0$, no network autocorrelation), spread *horizontally*. Its league mean climbed from below the time-invariant, season-agnostic baseline in the late 2000s to clearly above it by the early 2020s, the classic profile of a payoff-driven innovation copied among contemporaries. Because the expectation model excludes season, that drift is a horizontal-transmission signal rather than a modeling artifact.

This is where the study rejoins its predecessor. Mesoudi (2020) showed that a successful football tactic spreads horizontally, adopted by managers from their peers as its advantage becomes evident, and left vertical lineage transmission as open work. The NFL supplies the complementary half, and the two halves interlock. The trait that Mesoudi’s horizontal lens would track, the fitness-linked one, is exactly the trait our vertical lens finds absent from the coaching tree; the trait our vertical lens finds transmitting, schematic identity, is not the one that sweeps the league on a fitness gradient. Vertical transmission carries system identity; horizontal transmission carries the winning tactic. A

Two Fitnesses of NFL Head Coaches: Performance vs Progeny

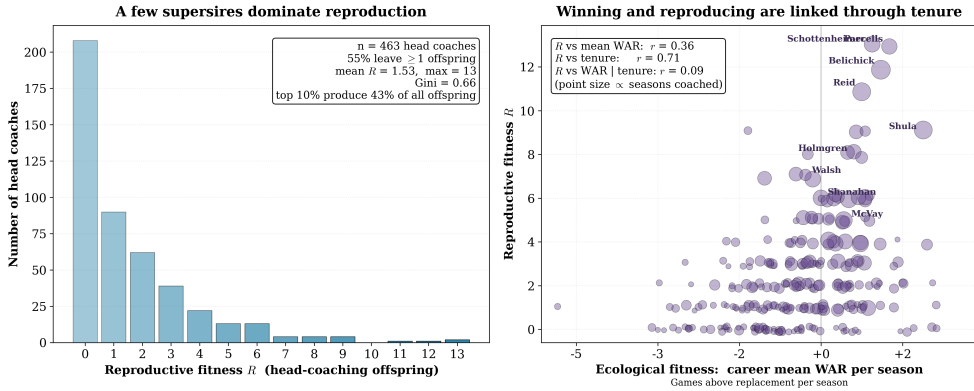


Figure 4. Two fitnesses of NFL head coaches. Left: the distribution of reproductive fitness (number of head-coaching proteges) is heavy-tailed (Gini 0.66; top decile produces 43% of all descendants). Right: ecological fitness (career mean WAR per season) versus reproductive fitness, point size proportional to seasons coached; the positive raw association ($r = 0.36$) is carried by tenure (offspring versus seasons coached $r = 0.71$), and net of tenure quality barely predicts offspring (partial $r = 0.09$).

study looking only down the tree would conclude aggression is absent from coaching culture; a study looking only at league-wide diffusion would miss that scheme is inherited. Read together, the two channels partition the strategic repertoire, and the era structure we remove as a statistical confounder is, substantively, the horizontal channel itself.

4.3 Defensive aggression: attitude and scheme entangled

Defensive aggression is the one composite that both transmits ($h^2 = 0.49$) and predicts winning ($S = 0.15$), and it is chiefly why the cross-trait relationship is not cleanly negative. It is best read not as a counterexample but as a reminder that our two-family taxonomy, approach versus identity, is cleaner on offense than on defense. “Aggression” on defense bundles two dimensions the offensive split holds apart. One is genuine attitude, how much risk a defense takes: loading the box, sending extra rushers. The other is schematic, which system a defense runs: man versus zone coverage is a system choice as much as operating from shotgun is, closer to a formation than to a fourth-down gamble. A composite that mixes an attitudinal dimension with a schematic one inherits some of the transmissibility of scheme and some of the fitness link of attitude, which is precisely the both-corner position it occupies. The sub-traits bear this out: box-stacking and pass-rush transmit strongly ($h^2 \approx 0.43$), like schematic identity, while man coverage transmits weakly ($h^2 = 0.14$); even within defensive aggression the components do not behave alike. The organizing distinction is therefore attitude versus scheme, not offense versus defense, and a cleaner future decomposition would separate a defense’s risk posture from its coverage system.

4.4 Coaching lineages transmit systems, not success

The pedigree logic that drives NFL hiring finds little support for its strongest claim. Working under a successful mentor does not, in these data, predict a protege’s own success (mentor–protege WAR resemblance near zero), and the in-game risk tendency that does predict success is not passed down. What coaching trees reliably transmit is schematic system, formation and tempo identity, and specifically through the coordinator who would install it. There is a labor-market reading of this asymmetry: an organization can observe and verify the system a coordinator runs far more readily than whether he will win, so hiring on visible schematic pedigree is hiring on the trait that transmits,

not the trait that drives value. What is legible (system) and what is decisive (in-game approach) come apart, and only the former leaves a lineage to point to. We offer this as interpretation rather than a tested claim; the confirmatory result is the decoupling itself.

4.5 Limitations

The transmission samples are small (about 40 offensive and 12–17 defensive coordinator-to-head-coach transitions), so the h^2 posteriors are wide and the cross-trait test in particular has low power; H3's wide interval should be read as inconclusive about a weak relationship, not as evidence for a positive one. Coaching pedigrees are non-tree, with multiple simultaneous mentors and overlapping careers, and coaching relatedness can only be assumed by analogy rather than derived from a Mendelian process, so we use a parent-offspring regression rather than a pedigree animal model. The genes attribute team play-calling to the head coach and do not isolate a coordinator's individual contribution; the transmission tests therefore ask whether the head-coaching style a coach apprenticed under predicts his own later style, a mentor-protege resemblance that is consistent with social learning but cannot on its own be separated from assortative hiring, in which head coaches recruit coordinators who already share their approach. We therefore read h^2 as vertical transmissibility in the broad sense. The defensive-aggression measures begin in 2016 and man coverage in 2018, and the defensive-aggression to WAR link may partly reflect reverse causation (better defensive talent enables man coverage and extra rushers). Finally, single-season WAR is noisy; we address this with inverse-variance weighting and treat the season-level estimates as conservative floors.

4.6 Conclusion

Treating the NFL coaching tree as a cultural pedigree, we find that the strategic traits that transmit are largely not the traits that win. Schematic identity is inherited but neutral; offensive aggression is selected but personal and, strikingly, not heritable despite being a stable individual trait; defensive aggression alone both transmits and wins. The negative heritability-fitness relationship of biological populations does not appear; transmission and selection are instead decoupled, and the fitness-linked behavior spreads horizontally rather than down the lineage. Coaching trees, in the end, propagate systems, not success.

Supplementary material

Table 5. Within-era phylogenetic signal on the mentor network (Moran's I on a row-normalized, undirected adjacency of mentor-protege ties; one-sided permutation p from within-career-midpoint-era label shuffles). $E[I] = -1/(N - 1)$ is the no-signal expectation. Under the exploratory Benjamini-Hochberg correction the defensive-aggression and tempo signals survive.

Gene	I	$E[I]$	z	p	N	edges
Offensive aggression	−0.01	−0.01	−0.04	0.52	139	440
Shotgun	+0.08	−0.01	+1.77	0.04	139	440
Tempo	+0.12	−0.01	+2.63	0.008	139	440
Defensive aggression	+0.21	−0.01	+3.03	0.002	95	216

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Competing Interests The author declares none.

Data Availability Statement The confirmatory analyses (C1–C3, hypotheses H1–H3) were preregistered before any confirmatory quantity was computed on the final measurement pipeline. All data and analysis code are public and seed-fixed, and the full pipeline runs from a single command; the preregistration, data, and code repository links are withheld here to preserve author anonymity and will be provided on acceptance.

Ethical Standards The research uses only publicly available data on professional-sport outcomes and involves no human or animal subjects; it meets all applicable ethical guidelines.

Author Contributions J.W. conceived the study, developed the methods and software, conducted the analysis, produced the visualizations, and wrote the manuscript.

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